

MycoGrow: A Standalone Browser-Based Platform for Fungal Radial Growth Analysis, Digital Phenotyping, and Comparative Statistics

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Abstract

Quantitative analysis of fungal radial growth is a cornerstone methodology in mycology, yet researchers continue to rely on manual measurements transcribed into spreadsheet software for statistical modeling—a workflow that is error-prone, time-consuming, and inaccessible in resource-limited settings. We present *MycoGrow*, an open-source, standalone HTML5 application that integrates the complete fungal growth analysis pipeline within a single browser-based interface. MycoGrow supports multi-isolate tracking (up to 50 simultaneous isolates), automated linear regression modeling with real-time visualization, a digital caliper for image-based colony measurement, quantitative colorimetry for objective phenotypic documentation, daily morphological logging, colony-forming unit (CFU) calculations, and comparative group statistics with bar chart generation. All computations are performed client-side using vanilla JavaScript, ensuring complete data privacy, zero-dependency operation, and full offline functionality. The application persists experimental data via browser `localStorage`, providing session continuity without server infrastructure. MycoGrow is freely available as a single HTML file and can be executed on any device with a modern web browser, making it particularly suitable for field laboratories and institutions in developing regions. This paper describes the software architecture, mathematical models, feature set, and potential applications of MycoGrow in streamlining mycological research workflows.

Keywords: Fungal growth rate, Radial growth, Linear regression, Colony morphology, Digital phenotyping, Colorimetry, Open-source software, HTML5, Offline laboratory tools.

1 Introduction

The quantitative measurement of fungal colony expansion on solid agar media remains one of the most fundamental and widely employed methodologies in contemporary mycological research [9, 8]. Radial growth rate, typically expressed in millimeters per day (mm/day), serves as a critical physiological metric for evaluating the effects of temperature, pH, water activity, nutrient availability, and antifungal compounds on filamentous fungi across diverse ecological and clinical contexts [4]. In standard laboratory practice, mycologists inoculate agar plates centrally, incubate them under controlled conditions, and measure colony diameters at regular time intervals using manual calipers or rulers. These raw measurements are subsequently transcribed into spreadsheet software such as Microsoft Excel, where linear regression analysis is performed to derive the radial growth rate from the slope of the colony radius versus time relationship [6]. This multi-step pipeline, while conceptually straightforward, introduces cumulative sources of human error at each stage of data transcription, mathematical computation, and graphical visualization, thereby compromising the reproducibility and efficiency of mycological investigations.

Beyond growth kinetics, the morphological characterization of fungal colonies constitutes an equally important dimension of mycological documentation. Colony color—both the obverse (forward) surface and the reverse—is a primary taxonomic descriptor used in the identification of species within genera such as *Aspergillus*, *Penicillium*, and *Trichoderma* [10]. Furthermore, changes in colony pigmentation over the course of incubation can reflect shifts in secondary metabolite production, sporulation status, or environmental stress responses [?]. Despite its taxonomic and physiological significance, colony color documentation remains overwhelmingly subjective, relying on descriptive terms such as “pale green,” “olive,” or “buff” that vary substantially between observers and lack quantitative precision. The absence of standardized, objective color measurement tools in routine mycological workflows represents a significant gap in phenotypic data quality that undermines the comparability of results across laboratories and studies.

The computational landscape available to mycologists for addressing these analytical challenges is paradoxically both abundant and inaccessible. Sophisticated statistical environments such as R, Python, and SAS can perform complex growth modeling, while proprietary image analysis software packages offer automated colony measurement capabilities [7]. However, these solutions impose substantial barriers to adoption, including steep learning curves, software licensing costs, dependency management complexities, and, in the case of cloud-based alternatives, mandatory internet connectivity [5]. For researchers operating in resource-limited academic institutions, field stations, or clinical laboratories in developing regions—where internet access may be intermittent and institutional software budgets constrained—these barriers effectively exclude them from the computational tools necessary for rigorous, modern mycological analysis [5, 5].

To address these converging challenges, we developed *MycoGrow*, an open-source, standalone HTML5 application that consolidates the entire fungal growth analysis pipeline—from raw measurement input through statistical modeling, phenotypic documentation, image-based measurement, and comparative group analysis—within a single, dependency-free browser interface. *MycoGrow* operates entirely client-side using vanilla JavaScript and the lightweight Chart.js library for visualization, requiring no server infrastructure, internet connectivity, software installation, or programming expertise. The application supports the simultaneous management of up to 50 fungal isolates as a digital laboratory notebook, with automatic data persistence via browser `localStorage`. Key innovations

include a built-in digital caliper for measuring colony diameters directly from uploaded Petri dish photographs, a quantitative colorimetry module that extracts precise hexadecimal color codes from colony images using the HTML5 Canvas API, and an integrated comparative statistics module that computes group-level growth rate comparisons with visualization. By encapsulating these capabilities within a single HTML file that can be distributed via email, USB drive, or institutional repository, MycoGrow aims to democratize access to quantitative mycological analysis tools and establish a new standard for portable, privacy-preserving laboratory software.

2 Software Architecture and Design

2.1 Standalone Client-Side Architecture

MycoGrow adheres to a strictly serverless, single-page application (SPA) architectural paradigm in which the entirety of the application logic, user interface, and data management subsystems are encapsulated within a single HTML file [3]. Unlike conventional web applications that depend on backend servers for computation, database management, and API orchestration, MycoGrow executes all operations exclusively within the user’s web browser via the JavaScript runtime environment. This design ensures that the application can be launched directly from a local file system (via the `file://` protocol) without requiring a web server, containerized deployment, or any network connectivity whatsoever. The single-file distribution model eliminates traditional software deployment barriers such as dependency resolution, version conflicts, and platform-specific compilation, allowing the tool to be disseminated through the simplest possible channels including email attachments, USB storage devices, or institutional file repositories.

2.2 Technology Stack

The application’s user interface is constructed using HTML5 semantic elements styled with the Tailwind CSS framework, loaded via a Content Delivery Network (CDN) link for initial setup but cached by the browser for subsequent offline use [11]. The responsive design system employs a grid-based layout that adapts seamlessly across desktop monitors, laboratory tablets, and mobile devices, ensuring usability across the heterogeneous hardware configurations encountered in research settings. Interactive data visualization is provided by Chart.js, a lightweight, canvas-based charting library that renders publication-quality scatter plots and bar charts without the overhead of larger visualization frameworks [1]. All computational logic—including linear regression, statistical analysis, pixel-level image processing, and state management—is implemented in vanilla JavaScript (ES6+), deliberately avoiding framework dependencies (e.g., React, Vue) to minimize the application’s complexity footprint and maximize long-term maintainability [2].

2.3 Data Persistence and Privacy

A distinguishing architectural feature of MycoGrow is its implementation of persistent data storage using the browser’s `localStorage` API. All experimental data—including isolate metadata, daily growth measurements, phenotypic color records, and group assignments—are automatically serialized to JSON and saved to `localStorage` upon every user inter-

action. This auto-save mechanism provides session continuity equivalent to desktop applications, allowing researchers to close and reopen the browser without data loss. Critically, because `localStorage` data resides exclusively on the user’s local machine and is never transmitted over any network, the architecture provides an inherent guarantee of complete data privacy and intellectual property protection [5]. This characteristic is particularly valuable when working with unpublished strain data, proprietary isolate collections, or clinical specimens subject to biosecurity regulations. A dedicated reset function allows users to clear all stored data and initiate a fresh experimental session when needed.

2.4 Print-Optimized Report Generation

MycoGrow incorporates a comprehensive `@media print` CSS layer that transforms the interactive interface into a clean, publication-ready document when the user invokes the browser’s native print dialog. During print rendering, all interactive controls, navigation elements, and file upload interfaces are automatically hidden, while the data entry tables, growth curve charts, and result cards are reformatted to fit standard A4 page dimensions with appropriate margins and typography. This approach leverages the browser’s built-in PDF rendering engine to produce vector-quality text and graphics, yielding output suitable for inclusion as supplementary material in peer-reviewed publications without requiring external PDF generation libraries.

3 Feature Set and Functional Description

MycoGrow provides an integrated suite of analytical modules designed to address the full spectrum of tasks encountered during fungal growth studies. This section describes each functional component in detail.

3.1 Multi-Isolate Laboratory Notebook

The application maintains a structured digital notebook capable of managing up to 50 fungal isolates simultaneously within a single experimental session. Each isolate record stores a comprehensive metadata profile including: (i) a user-defined isolate name or laboratory code, (ii) the isolation source (e.g., soil, plant tissue, clinical specimen), (iii) a categorical classification of the isolate type (Endophyte, Rhizosphere, Pathogen, Soil Saprophyte, or Other), and (iv) an experimental group assignment (e.g., Control, Treatment 1, Treatment 2) for subsequent comparative statistical analysis. Users navigate between isolates via a dropdown selector and can add or delete isolate records dynamically. This multi-isolate architecture transforms the tool from a single-use calculator into a persistent laboratory companion that accompanies the researcher throughout the duration of a growth study.

3.2 Daily Growth Data Entry with Phenotypic Logging

For each isolate, MycoGrow presents a tabular data entry interface where the user records daily observations. Each row corresponds to a single time point and captures the following parameters:

- **Day:** The incubation day (integer or decimal).

- **D1 and D2:** Two orthogonal colony diameter measurements (mm), from which the arithmetic mean is computed as the representative colony diameter for that time point.
- **Forward Color:** The color of the colony obverse surface.
- **Reverse Color:** The color of the colony reverse surface.
- **Include:** A boolean toggle that controls whether the data point is included in the linear regression calculation, enabling the user to selectively exclude lag-phase or stationary-phase measurements.

This per-day phenotypic color logging enables researchers to construct a temporal morphological profile for each isolate, documenting the progression of pigment changes across the incubation period—a feature absent from conventional spreadsheet-based workflows.

3.3 Digital Caliper: Image-Based Colony Measurement

MycroGrow incorporates a built-in digital caliper module that enables researchers to measure colony diameters directly from uploaded Petri dish photographs without requiring physical calipers. The measurement workflow consists of three steps:

1. **Image Upload:** The user uploads a photograph of the Petri dish. The image is rendered on an HTML5 `<canvas>` element at its native resolution.
2. **Calibration:** The user clicks on two diametrically opposite edges of the Petri dish to establish a reference distance. Given a known dish diameter (default: 90 mm), the system computes a pixels-per-millimeter conversion factor.
3. **Measurement:** The user clicks on two edges of the fungal colony. The Euclidean pixel distance is converted to millimeters using the calibration factor, and the result is displayed.

The measured value can be inserted directly into the data entry table with a single button click, eliminating transcription errors. All coordinate transformations account for CSS scaling between the canvas native resolution and its rendered display size.

3.4 Quantitative Colorimetry

A novel feature of MycoGrow is its quantitative colorimetry module, which replaces subjective verbal color descriptions with precise, reproducible hexadecimal (HEX) color codes extracted directly from colony images. When the user activates “Extract Color” mode and clicks on any point within the colony image, the application reads the RGB pixel values at the clicked coordinate using the Canvas API’s `getImageData()` method. These RGB values are converted to a HEX code (e.g., `#4A7C59`) and displayed alongside a visual color swatch. The extracted color can then be assigned to either the Forward or Reverse color field for the current day’s record. When a HEX code is stored in a color field, the data entry table renders a small colored square adjacent to the text, providing immediate visual feedback. This digital phenotyping capability transforms colony color documentation from a qualitative, observer-dependent process into a quantitative, reproducible measurement, enhancing the comparability of morphological data across studies and laboratories.

3.5 Colony-Forming Unit (CFU) Calculator

A supplementary CFU calculator is integrated into the application to support enumeration studies conducted in parallel with growth rate experiments. The user inputs the number

of colonies counted on a plate, the dilution factor exponent, and the plated volume (mL). The total count is calculated using the standard formula:

$$\text{CFU/mL (or g)} = \frac{\text{Colony Count} \times 10^{\text{dilution exponent}}}{\text{Plated Volume (mL)}} \quad (1)$$

The result is displayed in scientific notation for clarity and interpretability.

3.6 Comparative Group Statistics

MycoGrow includes an integrated comparative analysis module that enables researchers to compare growth rates across experimental groups without exporting data to external statistical software. When the user clicks “Compare Groups,” the application:

1. Groups all isolates by their assigned experimental group (or isolate type if no group is specified).
2. Computes the linear regression slope (growth rate) for each isolate.
3. Calculates the arithmetic mean growth rate for each group.
4. Renders a summary table displaying the group name, number of isolates, and mean growth rate.
5. Generates a grouped bar chart using Chart.js for visual comparison.

This module is presented within a modal overlay to avoid disrupting the primary data entry workflow and provides an immediate, at-a-glance comparison suitable for preliminary experimental assessment.

3.7 Data Export

All experimental data can be exported as a comprehensive CSV (Comma-Separated Values) file containing, for each data point across all isolates: the isolate name, isolation source, isolate type, day, D1, D2, average diameter, forward color, reverse color, inclusion status, calculated growth rate, R-squared value, and maximum diameter. This structured export format is directly compatible with downstream statistical analysis in R, Python, SPSS, or any spreadsheet application.

4 Mathematical Models

4.1 Linear Regression for Radial Growth Rate

The core analytical engine of MycoGrow employs ordinary least squares (OLS) linear regression to model the relationship between incubation time and colony diameter during the linear growth phase. Following the classical model of Trinci [9], the radial extension of a filamentous fungal colony on solid agar media, after an initial lag phase, proceeds at a constant rate that can be described by the linear equation:

$$D(t) = \alpha + \beta \cdot t \quad (2)$$

where $D(t)$ is the mean colony diameter (mm) at time t (days), α is the y-intercept representing the theoretical initial diameter, and β is the radial growth rate (mm/day) corresponding to the slope of the regression line.

For a set of n included data points $\{(t_1, D_1), (t_2, D_2), \dots, (t_n, D_n)\}$, the slope β is computed using the standard OLS formula:

$$\beta = \frac{n \sum_{i=1}^n t_i D_i - (\sum_{i=1}^n t_i) (\sum_{i=1}^n D_i)}{n \sum_{i=1}^n t_i^2 - (\sum_{i=1}^n t_i)^2} \quad (3)$$

and the intercept α is calculated as:

$$\alpha = \frac{\sum_{i=1}^n D_i - \beta \sum_{i=1}^n t_i}{n} \quad (4)$$

The quality of the linear fit is assessed using the coefficient of determination (R^2), which quantifies the proportion of variance in the dependent variable explained by the linear model:

$$R^2 = 1 - \frac{SS_{\text{res}}}{SS_{\text{tot}}} = 1 - \frac{\sum_{i=1}^n (D_i - \hat{D}_i)^2}{\sum_{i=1}^n (D_i - \bar{D})^2} \quad (5)$$

where $\hat{D}_i = \alpha + \beta t_i$ is the predicted diameter, and $\bar{D}$ is the mean of the observed diameters. An R^2 value approaching 1.0 indicates excellent linearity, confirming that the selected data points correspond to the linear growth phase. Users can selectively exclude data points from the lag or stationary phases using the “Include” toggle to optimize the regression fit, a critical feature for accurate growth rate determination since the linear model is biologically valid only during the exponential radial extension phase.

4.2 Diameter Averaging

For each time point, the mean colony diameter is computed from two orthogonal measurements (D_1 and D_2) to account for the frequently asymmetric radial expansion of fungal colonies:

$$D_{\text{avg}} = \frac{D_1 + D_2}{2} \quad (6)$$

If only a single measurement is provided (i.e., one of the two fields is empty or zero), the available measurement is used directly without averaging.

4.3 Image-Based Distance Calculation

The digital caliper module computes inter-point distances from canvas pixel coordinates using the Euclidean distance formula:

$$d_{\text{px}} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \quad (7)$$

During calibration, the pixels-per-millimeter conversion factor (k) is established as:

$$k = \frac{d_{\text{ref(px)}}}{D_{\text{ref(mm)}}} \quad (8)$$

where $d_{\text{ref(px)}}$ is the measured pixel distance across the Petri dish and $D_{\text{ref(mm)}}$ is the known dish diameter (typically 90 mm). Subsequent colony measurements are converted using:

$$D_{\text{colony}} = \frac{d_{\text{colony(px)}}}{k} \quad (9)$$

4.4 Colorimetric Conversion

The quantitative colorimetry module extracts pixel data from the HTML5 Canvas as RGBA values and converts the RGB components to hexadecimal notation using the standard transformation:

$$\text{HEX} = \# \text{hex}(R) \text{hex}(G) \text{hex}(B) \quad (10)$$

where each channel value (0–255) is converted to a two-digit hexadecimal string. This provides a universally standardized, device-independent color representation suitable for quantitative comparison across experiments and publications.

5 Use Cases and Applications

5.1 Antifungal Susceptibility Screening

Mycogrow’s multi-isolate notebook and comparative statistics module are particularly well-suited for antifungal susceptibility studies, where researchers must quantify the inhibitory effects of chemical or biological agents on fungal growth across multiple treatments and replicates. By assigning isolates to experimental groups (e.g., “Control,” “Fluconazole 10 $\mu\text{g}/\text{mL}$,” “Fluconazole 50 $\mu\text{g}/\text{mL}$ ”), the user can obtain immediate comparative bar charts of mean growth rates, facilitating rapid identification of effective concentrations without post-hoc analysis in external software. The selective data point exclusion feature allows researchers to focus the regression exclusively on the linear growth phase, ensuring that growth rate comparisons reflect genuine biological differences rather than artifacts of variable lag phase durations.

5.2 Taxonomic Colony Characterization

The integration of daily phenotypic color logging with quantitative colorimetry provides a structured framework for documenting the morphological progression of fungal isolates during taxonomic identification workflows. Rather than recording subjective color descriptions in a laboratory notebook, researchers can assign precise HEX color codes extracted directly from standardized colony photographs, creating a quantitative morphological dataset that is reproducible across observers and amenable to computational clustering or phylogenetic correlation analyses. The temporal color tracking capability—documenting forward and reverse colors at each time point—captures the dynamic pigmentation changes that are diagnostically significant in genera such as *Aspergillus*, *Fusarium*, and *Trichoderma*.

5.3 Environmental and Agricultural Mycology

Field mycologists studying soil fungal communities, rhizosphere associations, or endophytic colonization patterns frequently process large numbers of isolates with limited computational infrastructure. MycoGrow’s offline operation, single-file portability, and support for 50 simultaneous isolates make it an ideal companion for extended field campaigns where specimens must be cataloged, measured, and analyzed without reliable internet or access to institutional computing resources. The “Isolation Source” meta-data field enables downstream filtering and organization of isolates by ecological origin,

while the CSV export function provides a standardized data handoff point for subsequent community-level analyses.

5.4 Educational Applications

MycoGrow serves as an effective pedagogical tool for undergraduate and graduate mycology courses, providing students with an intuitive, visual interface for understanding the mathematical foundations of growth rate determination. The real-time chart rendering, which updates immediately as students modify data points or toggle the inclusion of lag-phase measurements, offers an interactive demonstration of how data selection influences regression parameters—a concept that is difficult to convey through static textbook presentations. The zero-installation deployment model allows instructors to distribute the tool to an entire class by sharing a single HTML file, eliminating the IT infrastructure requirements that often hinder the adoption of computational tools in teaching laboratories.

6 Conclusion and Future Directions

We have presented MycoGrow, a comprehensive, open-source, standalone HTML5 application that integrates multiple analytical functionalities essential to mycological research within a single, dependency-free browser-based platform. By consolidating linear regression growth modeling, image-based colony measurement, quantitative colorimetry, multi-isolate data management, and comparative group statistics into an offline-capable, privacy-preserving tool, MycoGrow addresses a persistent accessibility gap in the computational tools available to mycologists, particularly those working in resource-constrained environments.

The application’s single-file architecture, automatic data persistence via `localStorage`, and responsive design ensure that it can be deployed and utilized on virtually any device with a modern web browser, from high-specification laboratory workstations to basic tablets used in field settings. The quantitative colorimetry module represents a novel contribution to digital phenotyping in mycology, offering a standardized, reproducible alternative to the subjective color descriptions that currently prevail in taxonomic and morphological studies. Similarly, the integrated comparative statistics module reduces the dependence on external statistical software for preliminary experimental assessments, enabling researchers to evaluate treatment effects during the course of an experiment rather than exclusively at its conclusion.

Future development of MycoGrow will focus on several enhancements. First, the implementation of polynomial regression and logistic growth curve fitting to model non-linear growth phases and estimate carrying capacity. Second, the integration of computer vision algorithms (e.g., edge detection via `OpenCV.js`) for automated colony boundary recognition, eliminating the need for manual click-based measurement. Third, the addition of a predictive growth modeling module that extrapolates colony expansion trajectories from early-stage measurements to forecast plate coverage timing, thereby assisting in sub-culture scheduling. Fourth, the incorporation of standard deviation and error bar rendering in the comparative statistics module, along with automated one-way ANOVA significance testing. Finally, the development of a file-based save/load system using JSON file export and import, providing an alternative to `localStorage` for long-term archival and inter-device data transfer.

Software Availability

MycoGrow is released as open-source software under the MIT License. The complete source code is available as a single HTML file and can be accessed at the project repository. No installation, registration, or internet connection is required for operation.

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